HVAC Design Review Package

10,000 sf single-storey office

Central VAV air handler, interior VAV terminals, series fan-powered terminals on the perimeter

Revision 2: perimeter private offices and corner conference rooms, Denver, Colorado exposures

Item	Value
Document date	2026-10-21
Status	Schematic design for independent engineering review. Not for construction.
Model files	Office-10k-VAV-Mechanical.ifc (IFC4, feet), CreateOfficeModel.py (Revit API script), design.json (design data), four CSV schedules, plan.svg, isometric.svg
Repository folder	bim/office-10k-vav on branch ccr-e367aed5-vq3i83 of jamescatlas-bot/jamie-atlas-rehab
How it was produced	The design is defined once in design.py (geometry, zones, airflows). Scripts generate the IFC, the Revit script data, the drawings, the schedules and this document from that one source, so every number here matches the model.

Purpose of this package

Everything that went into the model is written out here so that a mechanical engineer can check the basis, the arithmetic and the layout without opening the model. Section 9 lists the items the author considers most in need of a professional check.

Contents

1. Summary
2. Design basis and assumptions
3. Thermal zones and spaces
4. Equipment
5. Drawings
6. Duct schedule
7. Air terminal schedule
8. Checks performed on the model
9. Items for engineering review
10. Revision history
11. How the model was built

1. Summary

The building is a 100 ft by 100 ft, 10,000 sf single-storey office with a 9 ft ceiling and a 14 ft roof deck on a 25 ft column grid. The perimeter is fitted out as 15 ft deep private offices, five per facade, with a 20 ft by 20 ft conference room at each corner and a 5 ft corridor behind the rooms. The interior is open office around a 24 ft by 16 ft mechanical room. Plan north is up.

A single indoor VAV air handler, AHU-1, rated 10,000 cfm with 1,800 cfm minimum outside air, serves 4 interior single-duct VAV terminals and 12 series fan-powered terminals on the perimeter through a medium-pressure ring main. Design primary airflow is 8,830 cfm (3,970 cfm interior, 4,860 cfm perimeter), which with the 7,290 cfm of fan air delivered by the perimeter boxes gives 11,260 cfm of supply to the rooms through 44 ceiling diffusers. Return is by ceiling plenum through 36 grilles.

Airflows carry Denver, Colorado exposure factors: relative to a north zone, east 1.25, west 1.35 and south 1.50. Perimeter offices run from 1.0 cfm/sf (north) to 1.5 cfm/sf (south); conference rooms are occupancy-driven at 1.2 cfm/sf times the worse of their two exposures; interior open office is 0.9 cfm/sf.

Key quantities

Item	Value
Floor area	10,000 sf (zones sum to 10,000.0 sf exactly)
Rooms	20 private offices, 4 conference rooms, 4 interior open-office zones, 1 mechanical room (29 spaces)
AHU-1	10,000 cfm supply, 1,800 cfm OA, 25 tons cooling, 300 MBH heating, 2.5 in. w.g. ESP, 15 hp supply fan
Terminal units	4 VAV (cooling only) and 12 series fan-powered boxes with electric reheat
Design primary air	8,830 cfm (diversity 0.88 on the AHU rating)
Air terminals	44 supply diffusers 24x24 lay-in, 36 return grilles 24x24 egg crate, 1 OA louver 36x30
Ductwork	934.5 ft total: SA/medium 316 ft, RA/low 14 ft, OA/low 42 ft, SA/low 564 ft
Duct runs / segments in model	111 runs, 150 straight segments, 120 fittings

2. Design basis and assumptions

Every assumption used to generate the model is listed here. Items marked (check) are rule-of-thumb values that should be replaced by calculation before the design is used for anything beyond planning.

2.1 Site and climate

Item	Value
Location	Denver, Colorado, ASHRAE climate zone 5B (high altitude, strong solar, dry). (check: altitude correction to fan and coil capacities is not applied)
Orientation	Plan up is true north. Facades are named N, E, S, W accordingly.
Exposure factors (cooling airflow multiplier vs north)	North 1.00, East 1.25, West 1.35, South 1.50 (check: derived from typical peak solar gain ratios for 5B, not from a load calculation)
Glazing	40% window-to-wall assumed on all facades, not modeled geometrically (check)

2.2 Building

Item	Value
Footprint	100 ft x 100 ft, origin at the south-west corner, X east, Y north
Levels	Level 1 at 0 ft, Roof at 14 ft. Slab on grade 6 in. (top at 0), roof deck 6 in. (bottom at 14 ft)
Ceiling	2x4 lay-in acoustical ceiling at 9 ft 0 in. everywhere except the mechanical room; plenum from 9 ft to 14 ft
Structure	25 ft x 25 ft grid, grids A to E (X = 0, 25, 50, 75, 100) and 1 to 5 (Y = 0, 25, 50, 75, 100); 25 columns 12 in. x 12 in. placeholder
Exterior walls	12 in. placeholder, full height 14 ft, centred 6 in. inside the building line
Interior partitions	6 in. placeholder; mechanical room walls full height to the roof deck; office and conference partitions to the 9 ft ceiling (plenum continuous above them)
Mechanical room M-101	X 38 to 62 ft, Y 54 to 70 ft (24 ft x 16 ft, 384 sf), no ceiling
Perimeter rooms	15 ft deep from the exterior face. Offices 12 ft wide (180 sf) with left edges at 20, 32, 44, 56 and 68 ft along each facade. Conference rooms 20 ft x 20 ft (400 sf) at each corner
Corridor	5 ft wide behind the rooms (depth 15 to 20 ft), open to the interior office, counted with the adjacent interior zone
Not modeled	Doors, windows, lighting, plumbing, fire protection, electrical, structure beyond placeholder columns, roof drains, equipment pads, hangers, insulation, flex duct

2.3 Airflow basis

Item	Value
Perimeter office supply (fan air)	Floor area x exposure factor x 1.0 cfm/sf: north 180, east 225, west 245, south 270 cfm per 180 sf office (check)
Conference room supply	400 sf x 1.2 cfm/sf (occupancy and equipment) x the larger exposure factor of the two facades: NE 600, NW 650, SE 720, SW 720 cfm (check: based on about 12 to 16 occupants)
Interior open office	0.90 cfm/sf: NW and NE 1,008 sf at 905 cfm, SW and SE 1,200 sf at 1,080 cfm
Series fan-powered box ratio	Fan airflow = 1.5 x design primary airflow. Primary = fan / 1.5 rounded to 5 cfm. Induced plenum air = 2,430 cfm in total (check: ratio, and whether parallel boxes would be preferred)
Minimum airflow	30% of design primary for every box (check against ASHRAE 62.1 zone minimums and 90.1 reheat limits)
Electric reheat	25 Btuh/sf of zone area, rounded up to whole kW (3,412 Btuh/kW). (check: Denver heating load with 40% glazing; also confirms whether hot-water reheat is preferred)
Outside air	1,800 cfm, about 18% of supply. Basis: ASHRAE 62.1 office rates, 5 cfm/person + 0.06 cfm/sf, for roughly 20 office occupants, 48 conference seats and 40 open-office occupants, divided by a system ventilation efficiency of about 0.75, then rounded up (check: run the 62.1 multiple-zone calculation with the actual occupancy)
AHU rating	10,000 cfm against 8,830 cfm block primary air (diversity 0.88). 25 tons is about 400 cfm/ton on the rated airflow (check: block cooling load calculation)

Item	Value
Return air	8,200 cfm to the AHU (supply minus OA). Rooms return 11,250 cfm to the plenum; 2,430 cfm is re-induced by the fan-powered boxes and about 620 cfm is available for relief, toilet exhaust and exfiltration (check: relief and building pressurization are not modeled)

2.4 Duct design basis

Item	Value
Pressure classes	Medium pressure (2 in. w.g. class) from the AHU through the ring main to the terminal inlets. Low pressure (1 in. w.g. class) downstream of every terminal, and for the return and outside-air ducts
Sizing method	Velocity limited, rectangular sizes chosen from a standard list in 2 in. steps: medium pressure at or below 1,500 fpm, low pressure at or below 900 fpm. Ring legs are held at a 12x10 minimum. Round runouts at or below 800 fpm. Sizes reduce 2 ft after each tap where the remaining airflow allows a smaller duct (check: equal-friction or static-regain check, and acoustic check of the 1,400 to 1,500 fpm mains)
Duct elevation	All horizontal ductwork at 11 ft 6 in. centreline (bottom of the largest 36x24 duct at 10 ft 6 in., 1 ft 6 in. above the ceiling). Runouts drop vertically to the diffuser necks at 9 ft 6 in.
Fittings	Elbows at every direction change, transitions at size changes, bull-head tees where mains split, takeoffs for branch taps. In the IFC these are placeholder blocks; in Revit they come from the duct type's routing preferences
Terminal inlets	Round, sized for at or below 2,000 fpm at design primary air: 6 to 10 in. for the perimeter boxes, 10 in. for the VAVs
Diffusers	24x24 lay-in, one per office, two per conference room, four per interior VAV. Neck sized at or below 800 fpm: 6 in. up to 155 cfm, 8 in. up to 280 cfm, 10 in. up to 435 cfm
Return grilles	24x24 egg crate in the ceiling, one per office, two per conference room, eight in the open office. Each grille is assigned its room's supply airflow (check: door undercuts or transfer grilles are not needed because every room has its own plenum return)
Flexible duct	Not modeled. Runouts are drawn as rigid round duct from the header to the diffuser neck
Materials	Not specified. Assume galvanized steel, SMACNA construction, externally insulated supply

3. Thermal zones and spaces

Zones are numbered clockwise from the north-west corner. Each perimeter zone is one room; interior zones are the open office quadrants including the corridor behind the perimeter rooms. The supply column is the air delivered to the room (fan air for fan-powered zones).

Number	Name	Type	Terminal box	Area (sf)	Design supply (cfm)	cfm/sf
101	Conference NW	conference	FPB-NW	400	650	1.62
102	North Office 1	office	FPB-N1	180	180	1.0
103	North Office 2	office	FPB-N1	180	180	1.0

Number	Name	Type	Terminal box	Area (sf)	Design supply (cfm)	cfm/sf
104	North Office 3	office	FPB-N1	180	180	1.0
105	North Office 4	office	FPB-N2	180	180	1.0
106	North Office 5	office	FPB-N2	180	180	1.0
107	Conference NE	conference	FPB-NE	400	600	1.5
108	East Office 1	office	FPB-E1	180	225	1.25
109	East Office 2	office	FPB-E1	180	225	1.25
110	East Office 3	office	FPB-E1	180	225	1.25
111	East Office 4	office	FPB-E2	180	225	1.25
112	East Office 5	office	FPB-E2	180	225	1.25
113	Conference SE	conference	FPB-SE	400	720	1.8
114	South Office 1	office	FPB-S1	180	270	1.5
115	South Office 2	office	FPB-S1	180	270	1.5
116	South Office 3	office	FPB-S1	180	270	1.5
117	South Office 4	office	FPB-S2	180	270	1.5
118	South Office 5	office	FPB-S2	180	270	1.5
119	Conference SW	conference	FPB-SW	400	720	1.8
120	West Office 1	office	FPB-W1	180	245	1.36
121	West Office 2	office	FPB-W1	180	245	1.36
122	West Office 3	office	FPB-W1	180	245	1.36
123	West Office 4	office	FPB-W2	180	245	1.36
124	West Office 5	office	FPB-W2	180	245	1.36
125	Interior NW Open Office	interior	VAV-1	1,008	905	0.9
126	Interior NE Open Office	interior	VAV-2	1,008	905	0.9
127	Interior SW Open Office	interior	VAV-3	1,200	1,080	0.9
128	Interior SE Open Office	interior	VAV-4	1,200	1,080	0.9
M-101	Mechanical Room	mechanical	AHU-1	384	0	

Zone boundaries (plan coordinates in feet, X east, Y north):

Zone	Polygon vertices (X, Y)
101	(0, 100) (20, 100) (20, 80) (0, 80)
102	(20, 100) (32, 100) (32, 85) (20, 85)
103	(32, 100) (44, 100) (44, 85) (32, 85)
104	(44, 100) (56, 100) (56, 85) (44, 85)
105	(56, 100) (68, 100) (68, 85) (56, 85)
106	(68, 100) (80, 100) (80, 85) (68, 85)
107	(100, 100) (100, 80) (80, 80) (80, 100)
108	(100, 80) (100, 68) (85, 68) (85, 80)
109	(100, 68) (100, 56) (85, 56) (85, 68)
110	(100, 56) (100, 44) (85, 44) (85, 56)
111	(100, 44) (100, 32) (85, 32) (85, 44)
112	(100, 32) (100, 20) (85, 20) (85, 32)
113	(100, 0) (80, 0) (80, 20) (100, 20)

Zone	Polygon vertices (X, Y)
114	(80, 0) (68, 0) (68, 15) (80, 15)
115	(68, 0) (56, 0) (56, 15) (68, 15)
116	(56, 0) (44, 0) (44, 15) (56, 15)
117	(44, 0) (32, 0) (32, 15) (44, 15)
118	(32, 0) (20, 0) (20, 15) (32, 15)
119	(0, 0) (0, 20) (20, 20) (20, 0)
120	(0, 20) (0, 32) (15, 32) (15, 20)
121	(0, 32) (0, 44) (15, 44) (15, 32)
122	(0, 44) (0, 56) (15, 56) (15, 44)
123	(0, 56) (0, 68) (15, 68) (15, 56)
124	(0, 68) (0, 80) (15, 80) (15, 68)
125	(15, 50) (50, 50) (50, 54) (38, 54) (38, 70) (50, 70) (50, 85) (20, 85) (20, 80) (15, 80)
126	(50, 50) (85, 50) (85, 80) (80, 80) (80, 85) (50, 85) (50, 70) (62, 70) (62, 54) (50, 54)
127	(15, 20) (20, 20) (20, 15) (50, 15) (50, 50) (15, 50)
128	(50, 15) (80, 15) (80, 20) (85, 20) (85, 50) (50, 50)
M-101	(38, 54) (62, 54) (62, 70) (38, 70)

4. Equipment

4.1 Equipment schedule

Tag	Type	Zone	Primary / supply cfm	Fan cfm	Min cfm	Inlet (in)	Heat (kW)	Location X (ft)	Location Y (ft)	Elevation (ft)	Size LxWxH (ft)
AHU-1	AHU		10000					51	62	0	16x6x7
L-1	LOUVER		1800					58	99.5	11.5	1x3x2.5
VAV-1	VAV	125	905		271	10		30	54	11.5	3x2x1.25
VAV-2	VAV	126	905		271	10		70	54	11.5	3x2x1.25
VAV-3	VAV	127	1080		324	10		34	40	11.5	3x2x1.25
VAV-4	VAV	128	1080		324	10		66	40	11.5	3x2x1.25
FPB-N1	FPB	102, 103, 104	360	540	108	6	4	38.0	82.5	11.5	5x3.5x1.5
FPB-N2	FPB	105, 106	240	360	72	6	3	68.0	82.5	11.5	5x3.5x1.5
FPB-NW	FPB	101	435	650	130	8	3	24.0	82.5	11.5	5x3.5x1.5
FPB-E1	FPB	108, 109, 110	450	675	135	8	4	82.5	62.0	11.5	5x3.5x1.5
FPB-E2	FPB	111, 112	300	450	90	6	3	82.5	32.0	11.5	5x3.5x1.5
FPB-NE	FPB	107	400	600	120	8	3	82.5	76.0	11.5	5x3.5x1.5
FPB-S1	FPB	114, 115, 116	540	810	162	8	4	62.0	17.5	11.5	5x3.5x1.5
FPB-S2	FPB	117, 118	360	540	108	6	3	32.0	17.5	11.5	5x3.5x1.5
FPB-SE	FPB	113	480	720	144	8	3	76.0	17.5	11.5	5x3.5x1.5
FPB-W1	FPB	120, 121, 122	490	735	147	8	4	17.5	38.0	11.5	5x3.5x1.5
FPB-W2	FPB	123, 124	325	490	97	6	3	17.5	68.0	11.5	5x3.5x1.5
FPB-SW	FPB	119	480	720	144	8	3	17.5	24.0	11.5	5x3.5x1.5

4.2 Equipment descriptions

AHU-1: Indoor central station VAV AHU, DX or CHW coil, economizer, MERV-13 filtration. Location (51, 62, 0) ft, footprint 16 x 6 ft, height 7 ft, flow direction (1, 0), rotation 0 deg. Inlet at (43, 62, 0), outlet at (59, 62, 0).

L-1: Drainable OA intake louver 36x30 on north wall. Location (58, 99.5, 11.5) ft, footprint 1 x 3 ft, height 2.5 ft, flow direction (0, -1), rotation -90 deg. Inlet at (58, 100, 11.5), outlet at (58, 99, 11.5).

VAV-1: Single-duct pressure-independent VAV terminal, cooling only. Location (30, 54, 11.5) ft, footprint 3 x 2 ft, height 1.25 ft, flow direction (0, 1), rotation 90 deg. Inlet at (30, 52.5, 11.5), outlet at (30, 55.5, 11.5). Serves zone(s) 125.

VAV-2: Single-duct pressure-independent VAV terminal, cooling only. Location (70, 54, 11.5) ft, footprint 3 x 2 ft, height 1.25 ft, flow direction (0, 1), rotation 90 deg. Inlet at (70, 52.5, 11.5), outlet at (70, 55.5, 11.5). Serves zone(s) 126.

VAV-3: Single-duct pressure-independent VAV terminal, cooling only. Location (34, 40, 11.5) ft, footprint 3 x 2 ft, height 1.25 ft, flow direction (0, -1), rotation -90 deg. Inlet at (34, 41.5, 11.5), outlet at (34, 38.5, 11.5). Serves zone(s) 127.

VAV-4: Single-duct pressure-independent VAV terminal, cooling only. Location (66, 40, 11.5) ft, footprint 3 x 2 ft, height 1.25 ft, flow direction (0, -1), rotation -90 deg. Inlet at (66, 41.5, 11.5), outlet at (66, 38.5, 11.5). Serves zone(s) 128.

FPB-N1: Series fan-powered terminal, ECM fan, electric reheat, plenum induction inlet. Location (38, 82.5, 11.5) ft, footprint 5 x 3.5 ft, height 1.5 ft, flow direction (0, 1), rotation 90 deg. Inlet at (38, 80, 11.5), outlet at (38, 85, 11.5). Exposure N. Serves zone(s) 102, 103, 104.

FPB-N2: Series fan-powered terminal, ECM fan, electric reheat, plenum induction inlet. Location (68, 82.5, 11.5) ft, footprint 5 x 3.5 ft, height 1.5 ft, flow direction (0, 1), rotation 90 deg. Inlet at (68, 80, 11.5), outlet at (68, 85, 11.5). Exposure N. Serves zone(s) 105, 106.

FPB-NW: Series fan-powered terminal, ECM fan, electric reheat, plenum induction inlet. Location (24, 82.5, 11.5) ft, footprint 5 x 3.5 ft, height 1.5 ft, flow direction (-1, 0), rotation 180 deg. Inlet at (26.5, 82.5, 11.5), outlet at (21.5, 82.5, 11.5). Exposure NW. Serves zone(s) 101.

FPB-E1: Series fan-powered terminal, ECM fan, electric reheat, plenum induction inlet. Location (82.5, 62, 11.5) ft, footprint 5 x 3.5 ft, height 1.5 ft, flow direction (1, 0), rotation 0 deg. Inlet at (80, 62, 11.5), outlet at (85, 62, 11.5). Exposure E. Serves zone(s) 108, 109, 110.

FPB-E2: Series fan-powered terminal, ECM fan, electric reheat, plenum induction inlet. Location (82.5, 32, 11.5) ft, footprint 5 x 3.5 ft, height 1.5 ft, flow direction (1, 0), rotation 0 deg. Inlet at (80, 32, 11.5), outlet at (85, 32, 11.5). Exposure E. Serves zone(s) 111, 112.

FPB-NE: Series fan-powered terminal, ECM fan, electric reheat, plenum induction inlet. Location (82.5, 76, 11.5) ft, footprint 5 x 3.5 ft, height 1.5 ft, flow direction (0, 1), rotation 90 deg. Inlet at (82.5, 73.5, 11.5), outlet at (82.5, 78.5, 11.5). Exposure EN. Serves zone(s) 107.

FPB-S1: Series fan-powered terminal, ECM fan, electric reheat, plenum induction inlet. Location (62, 17.5, 11.5) ft, footprint 5 x 3.5 ft, height 1.5 ft, flow direction (0, -1), rotation -90 deg. Inlet at (62, 20, 11.5), outlet at (62, 15, 11.5). Exposure S. Serves zone(s) 114, 115, 116.

FPB-S2: Series fan-powered terminal, ECM fan, electric reheat, plenum induction inlet. Location (32, 17.5, 11.5) ft, footprint 5 x 3.5 ft, height 1.5 ft, flow direction (0, -1), rotation -90 deg. Inlet at (32, 20, 11.5), outlet at (32, 15, 11.5). Exposure S. Serves zone(s) 117, 118.

FPB-SE: Series fan-powered terminal, ECM fan, electric reheat, plenum induction inlet. Location (76, 17.5, 11.5) ft, footprint 5 x 3.5 ft, height 1.5 ft, flow direction (1, 0), rotation 0 deg. Inlet at (73.5, 17.5, 11.5), outlet at (78.5, 17.5, 11.5). Exposure SE. Serves zone(s) 113.

FPB-W1: Series fan-powered terminal, ECM fan, electric reheat, plenum induction inlet. Location (17.5, 38, 11.5) ft, footprint 5 x 3.5 ft, height 1.5 ft, flow direction (-1, 0), rotation 180 deg. Inlet at (20, 38, 11.5), outlet at (15, 38, 11.5). Exposure W. Serves zone(s) 120, 121, 122.

FPB-W2: Series fan-powered terminal, ECM fan, electric reheat, plenum induction inlet. Location (17.5, 68, 11.5) ft, footprint 5 x 3.5 ft, height 1.5 ft, flow direction (-1, 0), rotation 180 deg. Inlet at (20, 68, 11.5), outlet at (15, 68, 11.5). Exposure W. Serves zone(s) 123, 124.

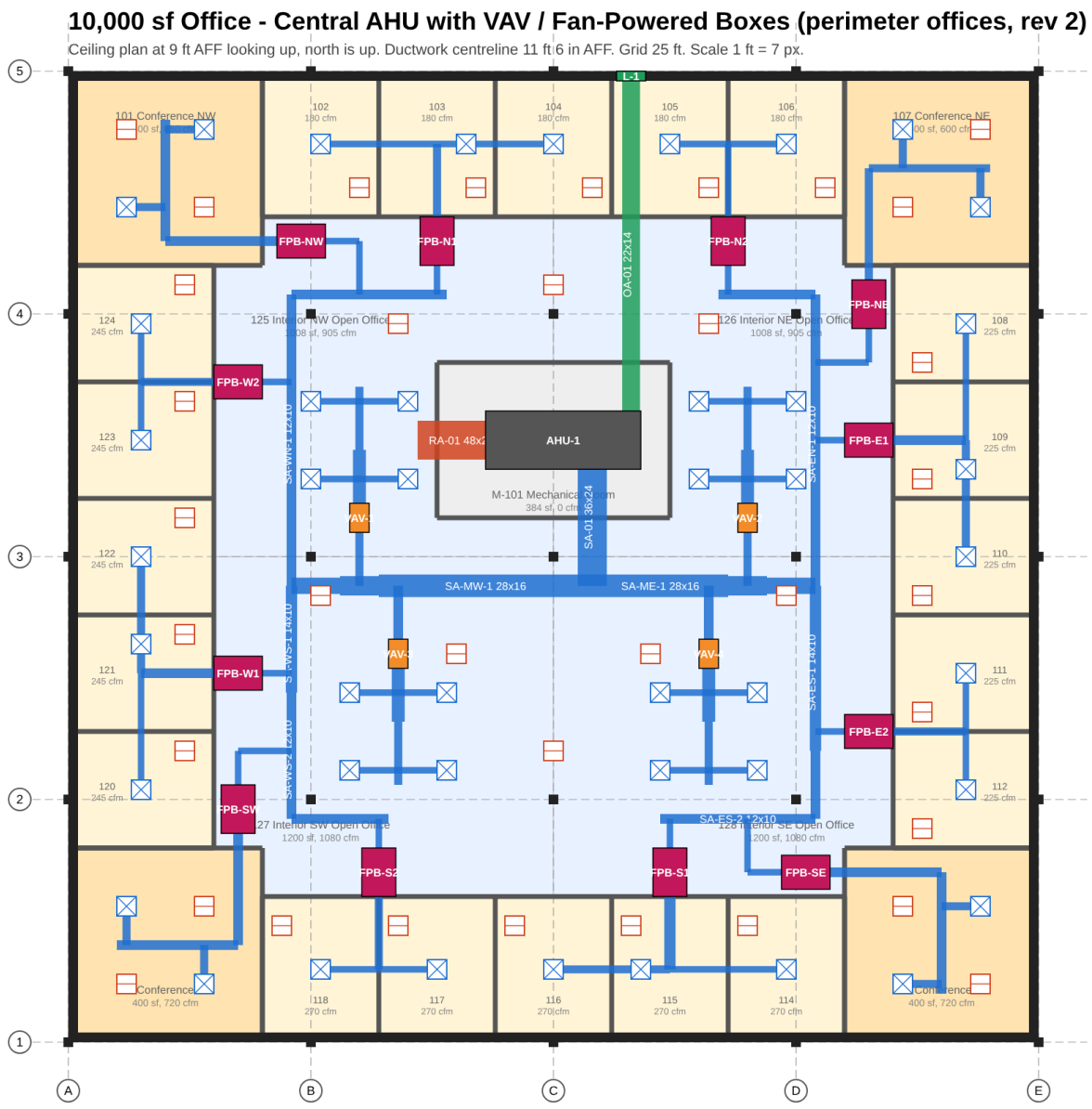
FPB-SW: Series fan-powered terminal, ECM fan, electric reheat, plenum induction inlet. Location (17.5, 24, 11.5) ft, footprint 5 x 3.5 ft, height 1.5 ft, flow direction (0, -1), rotation -90 deg. Inlet at (17.5, 26.5, 11.5), outlet at (17.5, 21.5, 11.5). Exposure WS. Serves zone(s) 119.

4.3 Notes on the equipment

- AHU-1 is a placeholder box of the right footprint (16 x 6 x 7 ft) with the return (48x28 at X = 45), outside air (22x14 at X = 58) and supply (36x24 at X = 54) connections all on top at 7 ft. A real unit will have end connections and needs the mechanical room layout, service clearance and a condensate drain checked.
- Fan-powered boxes are 5 x 3.5 x 1.5 ft placeholders mounted in the corridor ceiling with the primary inlet facing the ring main and the discharge toward the facade. The plenum induction inlet is on the side; no induction duct is drawn.
- VAV boxes are 3 x 2 x 1.25 ft placeholders in the open office ceiling.
- Electric reheat sizes are rounded up to whole kW from 25 Btuh/sf; the 3-office south boxes come to 4 kW, the conference boxes 3 kW.

5. Drawings

5.1 Ceiling plan at 9 ft, looking up

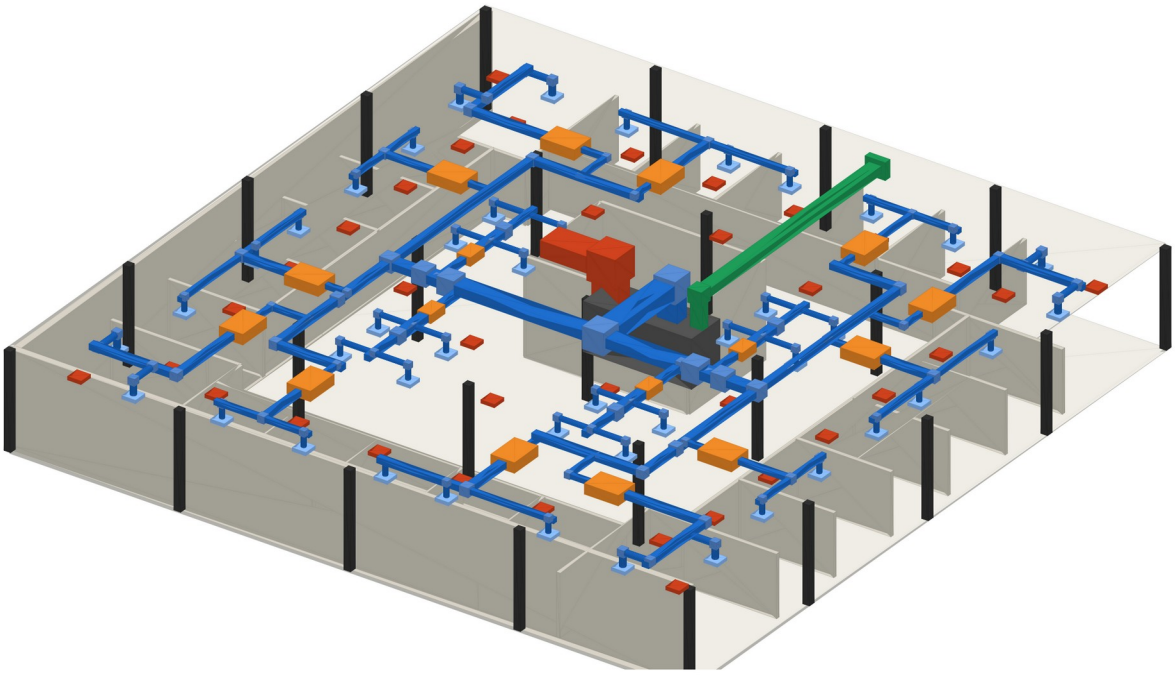


Blue: supply air. Red: return. Green: outside air. Orange: VAV boxes. Magenta: fan-powered boxes. Squares with an X are supply diffusers, squares with a bar are return grilles. Zone numbers and airflows are printed in each room.

5.2 Axonometric of the IFC model

Axonometric view of Office-10k-VAV-Mechanical.ifc (north and east walls and roof removed)

Blue: supply air ductwork and diffusers. Red: return. Green: outside air. Orange: VAV and fan-powered boxes. Grey: AHU, columns.



Roof, north wall and east wall removed. Partitions stop at the 9 ft ceiling; ductwork runs in the plenum above.

6. Duct schedule

Every run in the model. A run is a constant-size polyline; its interior vertices are elbows. 'Taps on' names the run a branch takes off from; 'From' and 'To' name equipment or terminals at the run ends. CAP marks a dead end. Points are centreline coordinates in feet (X, Y, Z), listed in the direction of airflow for supply and toward the AHU for return and outside air.

Run	Sys	Class	Description	Size	cfm	fpm	Len ft	Taps on	From	To	Centreline points (X, Y, Z) ft
SA-01	SA	medium	AHU-1 supply riser and main	36x24	8,830	1,472	19.5		AHU-1		(54, 62, 7) > (54, 62, 11.5) > (54, 47, 11.5)
SA-ME-1	SA	medium	Main east	28x16	4,395	1,413	14.0				(54, 47, 11.5) > (68, 47, 11.5)
SA-ME-2	SA	medium	Main east (reduced)	24x14	3,315	1,421	4.0				(68, 47, 11.5) > (72, 47, 11.5)
SA-ME-3	SA	medium	Main east (reduced)	20x12	2,410	1,446	5.0				(72, 47, 11.5) > (77, 47, 11.5)
SA-MW-1	SA	medium	Main west	28x16	4,435	1,426	22.0				(54, 47, 11.5) > (32, 47, 11.5)
SA-MW-2	SA	medium	Main west (reduced)	24x14	3,355	1,438	4.0				(32, 47, 11.5) > (28, 47, 11.5)
SA-MW-3	SA	medium	Main west (reduced)	20x12	2,450	1,470	5.0				(28, 47, 11.5) > (23, 47, 11.5)
SA-EN-1	SA	medium	East leg north to north facade	12x10	1,090	1,308	40.0			CAP	(77, 47, 11.5) > (77, 77, 11.5) > (67, 77, 11.5)
SA-ES-1	SA	medium	East leg south to south facade	14x10	1,320	1,358	17.0				(77, 47, 11.5) > (77, 30, 11.5)
SA-ES-2	SA	medium	East leg south to south facade (reduced)	12x10	1,020	1,224	23.0			CAP	(77, 30, 11.5) > (77, 23, 11.5) > (61, 23, 11.5)
SA-WN-1	SA	medium	West leg north to north facade	12x10	1,120	1,344	46.0			CAP	(23, 47, 11.5) > (23, 77, 11.5) > (39, 77, 11.5)
SA-WS-1	SA	medium	West leg south to south facade	14x10	1,330	1,368	11.0				(23, 47, 11.5) > (23, 36, 11.5)
SA-WS-2	SA	medium	West leg south to south facade (reduced)	12x10	840	1,008	23.0			CAP	(23, 36, 11.5) > (23, 23, 11.5) > (33, 23, 11.5)
RA-01	RA	low	Plenum return to AHU-1	48x28	8,200	879	13.5		PLENUM	AHU-1	(36, 62, 11.5) > (45, 62, 11.5) > (45, 62, 7)
OA-01	OA	low	Outside air from L-1 to AHU-1	22x14	1,800	842	41.5		L-1	AHU-1	(58, 99, 11.5) > (58, 62, 11.5) > (58, 62, 7)
SA-VAV-1	SA	medium	Tap to VAV-1	10x10	905	1,303	5.5	SA-MW-2		VAV-1	(30, 47, 11.5) > (30, 52.5, 11.5)
LP-VAV-1-1	SA	low	VAV-1 discharge header	16x10	905	815	5.5		VAV-1		(30, 55.5, 11.5) > (30, 61, 11.5)
LP-VAV-1-2	SA	low	VAV-1 discharge header (reduced)	10x8	452	814	6.5			CAP	(30, 61, 11.5) > (30, 67.5, 11.5)
LP-VAV-1-3	SA	low	Runout to SD-01	8 in. dia	225	645	7.0	LP-VAV-1-1		SD-01	(30, 58, 11.5) > (25, 58, 11.5) > (25, 58, 9.5)
LP-VAV-1-4	SA	low	Runout to SD-02	8 in. dia	225	645	7.0	LP-VAV-1-1		SD-02	(30, 58, 11.5) > (35, 58, 11.5) > (35, 58, 9.5)
LP-VAV-1-5	SA	low	Runout to SD-03	8 in. dia	225	645	7.0	LP-VAV-1-2		SD-03	(30, 66, 11.5) > (25, 66, 11.5) > (25, 66, 9.5)
LP-VAV-1-6	SA	low	Runout to SD-04	8 in. dia	225	645	7.0	LP-VAV-1-2		SD-04	(30, 66, 11.5) > (35, 66, 11.5) > (35, 66, 9.5)
SA-VAV-2	SA	medium	Tap to VAV-2	10x10	905	1,303	5.5	SA-ME-2		VAV-2	(70, 47, 11.5) > (70, 52.5, 11.5)

Run	Sys	Class	Description	Size	cfm	fpm	Len ft	Taps on	From	To	Centreline points (X, Y, Z) ft
LP-VAV-2-1	SA	low	VAV-2 discharge header	16x10	905	815	5.5		VAV-2		(70, 55.5, 11.5) > (70, 61, 11.5)
LP-VAV-2-2	SA	low	VAV-2 discharge header (reduced)	10x8	452	814	6.5			CAP	(70, 61, 11.5) > (70, 67.5, 11.5)
LP-VAV-2-3	SA	low	Runout to SD-05	8 in. dia	225	645	7.0	LP-VAV-2-1		SD-05	(70, 58, 11.5) > (65, 58, 11.5) > (65, 58, 9.5)
LP-VAV-2-4	SA	low	Runout to SD-06	8 in. dia	225	645	7.0	LP-VAV-2-1		SD-06	(70, 58, 11.5) > (75, 58, 11.5) > (75, 58, 9.5)
LP-VAV-2-5	SA	low	Runout to SD-07	8 in. dia	225	645	7.0	LP-VAV-2-2		SD-07	(70, 66, 11.5) > (65, 66, 11.5) > (65, 66, 9.5)
LP-VAV-2-6	SA	low	Runout to SD-08	8 in. dia	225	645	7.0	LP-VAV-2-2		SD-08	(70, 66, 11.5) > (75, 66, 11.5) > (75, 66, 9.5)
SA-VAV-3	SA	medium	Tap to VAV-3	12x10	1,080	1,296	5.5	SA-MW-1		VAV-3	(34, 47, 11.5) > (34, 41.5, 11.5)
LP-VAV-3-1	SA	low	VAV-3 discharge header	16x12	1,080	810	5.5		VAV-3		(34, 38.5, 11.5) > (34, 33, 11.5)
LP-VAV-3-2	SA	low	VAV-3 discharge header (reduced)	10x10	540	778	6.5			CAP	(34, 33, 11.5) > (34, 26.5, 11.5)
LP-VAV-3-3	SA	low	Runout to SD-09	8 in. dia	270	773	7.0	LP-VAV-3-1		SD-09	(34, 36, 11.5) > (29, 36, 11.5) > (29, 36, 9.5)
LP-VAV-3-4	SA	low	Runout to SD-10	8 in. dia	270	773	7.0	LP-VAV-3-1		SD-10	(34, 36, 11.5) > (39, 36, 11.5) > (39, 36, 9.5)
LP-VAV-3-5	SA	low	Runout to SD-11	8 in. dia	270	773	7.0	LP-VAV-3-2		SD-11	(34, 28, 11.5) > (29, 28, 11.5) > (29, 28, 9.5)
LP-VAV-3-6	SA	low	Runout to SD-12	8 in. dia	270	773	7.0	LP-VAV-3-2		SD-12	(34, 28, 11.5) > (39, 28, 11.5) > (39, 28, 9.5)
SA-VAV-4	SA	medium	Tap to VAV-4	12x10	1,080	1,296	5.5	SA-ME-1		VAV-4	(66, 47, 11.5) > (66, 41.5, 11.5)
LP-VAV-4-1	SA	low	VAV-4 discharge header	16x12	1,080	810	5.5		VAV-4		(66, 38.5, 11.5) > (66, 33, 11.5)
LP-VAV-4-2	SA	low	VAV-4 discharge header (reduced)	10x10	540	778	6.5			CAP	(66, 33, 11.5) > (66, 26.5, 11.5)
LP-VAV-4-3	SA	low	Runout to SD-13	8 in. dia	270	773	7.0	LP-VAV-4-1		SD-13	(66, 36, 11.5) > (61, 36, 11.5) > (61, 36, 9.5)
LP-VAV-4-4	SA	low	Runout to SD-14	8 in. dia	270	773	7.0	LP-VAV-4-1		SD-14	(66, 36, 11.5) > (71, 36, 11.5) > (71, 36, 9.5)
LP-VAV-4-5	SA	low	Runout to SD-15	8 in. dia	270	773	7.0	LP-VAV-4-2		SD-15	(66, 28, 11.5) > (61, 28, 11.5) > (61, 28, 9.5)
LP-VAV-4-6	SA	low	Runout to SD-16	8 in. dia	270	773	7.0	LP-VAV-4-2		SD-16	(66, 28, 11.5) > (71, 28, 11.5) > (71, 28, 9.5)
SA-FPB-N1	SA	medium	Tap to FPB-N1	8x8	360	810	3.0	SA-WN-1		FPB-N1	(38, 77, 11.5) > (38, 80, 11.5)
LP-FPB-N1-0	SA	low	FPB-N1 discharge header	10x10	540	778	7.5		FPB-N1		(38, 85, 11.5) > (38, 92.5, 11.5)
LP-FPB-N1-L	SA	low	FPB-N1 left leg	8x8	180	405	13.0			CAP	(38, 92.5, 11.5) > (25, 92.5, 11.5)
LP-FPB-N1-R	SA	low	FPB-N1 right leg	8x8	360	810	13.0			CAP	(38, 92.5, 11.5) > (51, 92.5, 11.5)
LP-FPB-N1-1	SA	low	Drop to SD-17	8 in. dia	180	516	2.0	LP-FPB-N1-L		SD-17	(26, 92.5, 11.5) > (26, 92.5, 9.5)
LP-FPB-N1-2	SA	low	Drop to SD-18	8 in. dia	180	516	2.0	LP-FPB-N1-R		SD-18	(41, 92.5, 11.5) > (41, 92.5, 9.5)
LP-FPB-N1-3	SA	low	Drop to SD-19	8 in. dia	180	516	2.0	LP-FPB-N1-R		SD-19	(50, 92.5, 11.5) > (50, 92.5, 9.5)

Run	Sys	Class	Description	Size	cfm	fpm	Len ft	Taps on	From	To	Centreline points (X, Y, Z) ft
SA-FPB-N2	SA	medium	Tap to FPB-N2	8x8	240	540	3.0	SA-EN-1		FPB-N2	(68, 77, 11.5) > (68, 80, 11.5)
LP-FPB-N2-0	SA	low	FPB-N2 discharge header	8x8	360	810	7.5		FPB-N2		(68, 85, 11.5) > (68, 92.5, 11.5)
LP-FPB-N2-L	SA	low	FPB-N2 left leg	8x8	180	405	7.0			CAP	(68, 92.5, 11.5) > (61, 92.5, 11.5)
LP-FPB-N2-R	SA	low	FPB-N2 right leg	8x8	180	405	7.0			CAP	(68, 92.5, 11.5) > (75, 92.5, 11.5)
LP-FPB-N2-1	SA	low	Drop to SD-20	8 in. dia	180	516	2.0	LP-FPB-N2-L		SD-20	(62, 92.5, 11.5) > (62, 92.5, 9.5)
LP-FPB-N2-2	SA	low	Drop to SD-21	8 in. dia	180	516	2.0	LP-FPB-N2-R		SD-21	(74, 92.5, 11.5) > (74, 92.5, 9.5)
SA-FPB-NW	SA	medium	Tap to FPB-NW	8x8	435	979	9.0	SA-WN-1		FPB-NW	(30, 77, 11.5) > (30, 82.5, 11.5) > (26.5, 82.5, 11.5)
LP-FPB-NW-0	SA	low	FPB-NW discharge header	12x10	650	780	24.0		FPB-NW	CAP	(21.5, 82.5, 11.5) > (10, 82.5, 11.5) > (10, 95, 11.5)
LP-FPB-NW-1	SA	low	Runout to SD-22	10 in. dia	325	596	6.0	LP-FPB-NW-0		SD-22	(10, 86, 11.5) > (6, 86, 11.5) > (6, 86, 9.5)
LP-FPB-NW-2	SA	low	Runout to SD-23	10 in. dia	325	596	6.0	LP-FPB-NW-0		SD-23	(10, 94, 11.5) > (14, 94, 11.5) > (14, 94, 9.5)
SA-FPB-E1	SA	medium	Tap to FPB-E1	8x8	450	1,013	3.0	SA-EN-1		FPB-E1	(77, 62, 11.5) > (80, 62, 11.5)
LP-FPB-E1-0	SA	low	FPB-E1 discharge header	12x10	675	810	7.5		FPB-E1		(85, 62, 11.5) > (92.5, 62, 11.5)
LP-FPB-E1-L	SA	low	FPB-E1 left leg	8x8	225	506	13.0			CAP	(92.5, 62, 11.5) > (92.5, 75, 11.5)
LP-FPB-E1-R	SA	low	FPB-E1 right leg	10x8	450	810	13.0			CAP	(92.5, 62, 11.5) > (92.5, 49, 11.5)
LP-FPB-E1-1	SA	low	Drop to SD-24	8 in. dia	225	645	2.0	LP-FPB-E1-L		SD-24	(92.5, 74, 11.5) > (92.5, 74, 9.5)
LP-FPB-E1-2	SA	low	Drop to SD-25	8 in. dia	225	645	2.0	LP-FPB-E1-R		SD-25	(92.5, 59, 11.5) > (92.5, 59, 9.5)
LP-FPB-E1-3	SA	low	Drop to SD-26	8 in. dia	225	645	2.0	LP-FPB-E1-R		SD-26	(92.5, 50, 11.5) > (92.5, 50, 9.5)
SA-FPB-E2	SA	medium	Tap to FPB-E2	8x8	300	675	3.0	SA-ES-1		FPB-E2	(77, 32, 11.5) > (80, 32, 11.5)
LP-FPB-E2-0	SA	low	FPB-E2 discharge header	10x8	450	810	7.5		FPB-E2		(85, 32, 11.5) > (92.5, 32, 11.5)
LP-FPB-E2-L	SA	low	FPB-E2 left leg	8x8	225	506	7.0			CAP	(92.5, 32, 11.5) > (92.5, 39, 11.5)
LP-FPB-E2-R	SA	low	FPB-E2 right leg	8x8	225	506	7.0			CAP	(92.5, 32, 11.5) > (92.5, 25, 11.5)
LP-FPB-E2-1	SA	low	Drop to SD-27	8 in. dia	225	645	2.0	LP-FPB-E2-L		SD-27	(92.5, 38, 11.5) > (92.5, 38, 9.5)
LP-FPB-E2-2	SA	low	Drop to SD-28	8 in. dia	225	645	2.0	LP-FPB-E2-R		SD-28	(92.5, 26, 11.5) > (92.5, 26, 9.5)
SA-FPB-NE	SA	medium	Tap to FPB-NE	8x8	400	900	9.0	SA-EN-1		FPB-NE	(77, 70, 11.5) > (82.5, 70, 11.5) > (82.5, 73.5, 11.5)
LP-FPB-NE-0	SA	low	FPB-NE discharge header	10x10	600	864	24.0		FPB-NE	CAP	(82.5, 78.5, 11.5) > (82.5, 90, 11.5) > (95, 90, 11.5)
LP-FPB-NE-1	SA	low	Runout to SD-29	10 in. dia	300	550	6.0	LP-FPB-NE-0		SD-29	(86, 90, 11.5) > (86, 94, 11.5) > (86, 94, 9.5)
LP-FPB-NE-2	SA	low	Runout to SD-30	10 in. dia	300	550	6.0	LP-FPB-NE-0		SD-30	(94, 90, 11.5) > (94, 86, 11.5) > (94, 86, 9.5)

Run	Sys	Class	Description	Size	cfm	fpm	Len ft	Taps on	From	To	Centreline points (X, Y, Z) ft
SA-FPB-S1	SA	medium	Tap to FPB-S1	8x8	540	1,215	3.0	SA-ES-2		FPB-S1	(62, 23, 11.5) > (62, 20, 11.5)
LP-FPB-S1-0	SA	low	FPB-S1 discharge header	14x10	810	833	7.5		FPB-S1		(62, 15, 11.5) > (62, 7.5, 11.5)
LP-FPB-S1-L	SA	low	FPB-S1 left leg	8x8	270	608	13.0			CAP	(62, 7.5, 11.5) > (75, 7.5, 11.5)
LP-FPB-S1-R	SA	low	FPB-S1 right leg	10x10	540	778	13.0			CAP	(62, 7.5, 11.5) > (49, 7.5, 11.5)
LP-FPB-S1-1	SA	low	Drop to SD-31	8 in. dia	270	773	2.0	LP-FPB-S1-L		SD-31	(74, 7.5, 11.5) > (74, 7.5, 9.5)
LP-FPB-S1-2	SA	low	Drop to SD-32	8 in. dia	270	773	2.0	LP-FPB-S1-R		SD-32	(59, 7.5, 11.5) > (59, 7.5, 9.5)
LP-FPB-S1-3	SA	low	Drop to SD-33	8 in. dia	270	773	2.0	LP-FPB-S1-R		SD-33	(50, 7.5, 11.5) > (50, 7.5, 9.5)
SA-FPB-S2	SA	medium	Tap to FPB-S2	8x8	360	810	3.0	SA-WS-2		FPB-S2	(32, 23, 11.5) > (32, 20, 11.5)
LP-FPB-S2-0	SA	low	FPB-S2 discharge header	10x10	540	778	7.5		FPB-S2		(32, 15, 11.5) > (32, 7.5, 11.5)
LP-FPB-S2-L	SA	low	FPB-S2 left leg	8x8	270	608	7.0			CAP	(32, 7.5, 11.5) > (39, 7.5, 11.5)
LP-FPB-S2-R	SA	low	FPB-S2 right leg	8x8	270	608	7.0			CAP	(32, 7.5, 11.5) > (25, 7.5, 11.5)
LP-FPB-S2-1	SA	low	Drop to SD-34	8 in. dia	270	773	2.0	LP-FPB-S2-L		SD-34	(38, 7.5, 11.5) > (38, 7.5, 9.5)
LP-FPB-S2-2	SA	low	Drop to SD-35	8 in. dia	270	773	2.0	LP-FPB-S2-R		SD-35	(26, 7.5, 11.5) > (26, 7.5, 9.5)
SA-FPB-SE	SA	medium	Tap to FPB-SE	8x8	480	1,080	9.0	SA-ES-2		FPB-SE	(70, 23, 11.5) > (70, 17.5, 11.5) > (73.5, 17.5, 11.5)
LP-FPB-SE-0	SA	low	FPB-SE discharge header	12x10	720	864	24.0		FPB-SE	CAP	(78.5, 17.5, 11.5) > (90, 17.5, 11.5) > (90, 5, 11.5)
LP-FPB-SE-1	SA	low	Runout to SD-36	10 in. dia	360	660	6.0	LP-FPB-SE-0		SD-36	(90, 14, 11.5) > (94, 14, 11.5) > (94, 14, 9.5)
LP-FPB-SE-2	SA	low	Runout to SD-37	10 in. dia	360	660	6.0	LP-FPB-SE-0		SD-37	(90, 6, 11.5) > (86, 6, 11.5) > (86, 6, 9.5)
SA-FPB-W1	SA	medium	Tap to FPB-W1	8x8	490	1,103	3.0	SA-WS-1		FPB-W1	(23, 38, 11.5) > (20, 38, 11.5)
LP-FPB-W1-0	SA	low	FPB-W1 discharge header	12x10	735	882	7.5		FPB-W1		(15, 38, 11.5) > (7.5, 38, 11.5)
LP-FPB-W1-L	SA	low	FPB-W1 left leg	8x8	245	551	13.0			CAP	(7.5, 38, 11.5) > (7.5, 25, 11.5)
LP-FPB-W1-R	SA	low	FPB-W1 right leg	10x8	490	882	13.0			CAP	(7.5, 38, 11.5) > (7.5, 51, 11.5)
LP-FPB-W1-1	SA	low	Drop to SD-38	8 in. dia	245	702	2.0	LP-FPB-W1-L		SD-38	(7.5, 26, 11.5) > (7.5, 26, 9.5)
LP-FPB-W1-2	SA	low	Drop to SD-39	8 in. dia	245	702	2.0	LP-FPB-W1-R		SD-39	(7.5, 41, 11.5) > (7.5, 41, 9.5)
LP-FPB-W1-3	SA	low	Drop to SD-40	8 in. dia	245	702	2.0	LP-FPB-W1-R		SD-40	(7.5, 50, 11.5) > (7.5, 50, 9.5)
SA-FPB-W2	SA	medium	Tap to FPB-W2	8x8	325	731	3.0	SA-WN-1		FPB-W2	(23, 68, 11.5) > (20, 68, 11.5)
LP-FPB-W2-0	SA	low	FPB-W2 discharge header	10x8	490	882	7.5		FPB-W2		(15, 68, 11.5) > (7.5, 68, 11.5)
LP-FPB-W2-L	SA	low	FPB-W2 left leg	8x8	245	551	7.0			CAP	(7.5, 68, 11.5) > (7.5, 61, 11.5)

Run	Sys	Class	Description	Size	cfm	fpm	Len ft	Taps on	From	To	Centreline points (X, Y, Z) ft
LP-FPB-W2-R	SA	low	FPB-W2 right leg	8x8	245	551	7.0			CAP	(7.5, 68, 11.5) > (7.5, 75, 11.5)
LP-FPB-W2-1	SA	low	Drop to SD-41	8 in. dia	245	702	2.0	LP-FPB-W2-L		SD-41	(7.5, 62, 11.5) > (7.5, 62, 9.5)
LP-FPB-W2-2	SA	low	Drop to SD-42	8 in. dia	245	702	2.0	LP-FPB-W2-R		SD-42	(7.5, 74, 11.5) > (7.5, 74, 9.5)
SA-FPB-SW	SA	medium	Tap to FPB-SW	8x8	480	1,080	9.0	SA-WS-2		FPB-SW	(23, 30, 11.5) > (17.5, 30, 11.5) > (17.5, 26.5, 11.5)
LP-FPB-SW-0	SA	low	FPB-SW discharge header	12x10	720	864	24.0		FPB-SW	CAP	(17.5, 21.5, 11.5) > (17.5, 10, 11.5) > (5, 10, 11.5)
LP-FPB-SW-1	SA	low	Runout to SD-43	10 in. dia	360	660	6.0	LP-FPB-SW-0		SD-43	(14, 10, 11.5) > (14, 6, 11.5) > (14, 6, 9.5)
LP-FPB-SW-2	SA	low	Runout to SD-44	10 in. dia	360	660	6.0	LP-FPB-SW-0		SD-44	(6, 10, 11.5) > (6, 14, 11.5) > (6, 14, 9.5)

7. Air terminal schedule

Supply diffusers (SD) and return grilles (RG), all 24x24 lay-in at the 9 ft ceiling. 'Served by' is the terminal box for diffusers; return grilles return to the plenum.

Tag	Type	Room	Served by	cfm	Face in.	Neck in.	X ft	Y ft	Ceiling ft
SD-01	Supply diffuser	125	VAV-1	225	24	8	25	58	9
SD-02	Supply diffuser	125	VAV-1	225	24	8	35	58	9
SD-03	Supply diffuser	125	VAV-1	225	24	8	25	66	9
SD-04	Supply diffuser	125	VAV-1	225	24	8	35	66	9
SD-05	Supply diffuser	126	VAV-2	225	24	8	65	58	9
SD-06	Supply diffuser	126	VAV-2	225	24	8	75	58	9
SD-07	Supply diffuser	126	VAV-2	225	24	8	65	66	9
SD-08	Supply diffuser	126	VAV-2	225	24	8	75	66	9
SD-09	Supply diffuser	127	VAV-3	270	24	8	29	36	9
SD-10	Supply diffuser	127	VAV-3	270	24	8	39	36	9
SD-11	Supply diffuser	127	VAV-3	270	24	8	29	28	9
SD-12	Supply diffuser	127	VAV-3	270	24	8	39	28	9
SD-13	Supply diffuser	128	VAV-4	270	24	8	61	36	9
SD-14	Supply diffuser	128	VAV-4	270	24	8	71	36	9
SD-15	Supply diffuser	128	VAV-4	270	24	8	61	28	9
SD-16	Supply diffuser	128	VAV-4	270	24	8	71	28	9
SD-17	Supply diffuser	102	FPB-N1	180	24	8	26	92.5	9
RG-01	Return grille	102	PLENUM	180	24		30	88	9
SD-18	Supply diffuser	103	FPB-N1	180	24	8	41	92.5	9

Tag	Type	Room	Served by	cfm	Face in.	Neck in.	X ft	Y ft	Ceiling ft
RG-02	Return grille	103	PLENUM	180	24		42	88	9
SD-19	Supply diffuser	104	FPB-N1	180	24	8	50	92.5	9
RG-03	Return grille	104	PLENUM	180	24		54	88	9
SD-20	Supply diffuser	105	FPB-N2	180	24	8	62	92.5	9
RG-04	Return grille	105	PLENUM	180	24		66	88	9
SD-21	Supply diffuser	106	FPB-N2	180	24	8	74	92.5	9
RG-05	Return grille	106	PLENUM	180	24		78	88	9
SD-22	Supply diffuser	101	FPB-NW	325	24	10	6	86	9
SD-23	Supply diffuser	101	FPB-NW	325	24	10	14	94	9
RG-06	Return grille	101	PLENUM	325	24		14	86	9
RG-07	Return grille	101	PLENUM	325	24		6	94	9
SD-24	Supply diffuser	108	FPB-E1	225	24	8	92.5	74	9
RG-08	Return grille	108	PLENUM	225	24		88	70	9
SD-25	Supply diffuser	109	FPB-E1	225	24	8	92.5	59	9
RG-09	Return grille	109	PLENUM	225	24		88	58	9
SD-26	Supply diffuser	110	FPB-E1	225	24	8	92.5	50	9
RG-10	Return grille	110	PLENUM	225	24		88	46	9
SD-27	Supply diffuser	111	FPB-E2	225	24	8	92.5	38	9
RG-11	Return grille	111	PLENUM	225	24		88	34	9
SD-28	Supply diffuser	112	FPB-E2	225	24	8	92.5	26	9
RG-12	Return grille	112	PLENUM	225	24		88	22	9
SD-29	Supply diffuser	107	FPB-NE	300	24	10	86	94	9
SD-30	Supply diffuser	107	FPB-NE	300	24	10	94	86	9
RG-13	Return grille	107	PLENUM	300	24		86	86	9
RG-14	Return grille	107	PLENUM	300	24		94	94	9
SD-31	Supply diffuser	114	FPB-S1	270	24	8	74	7.5	9
RG-15	Return grille	114	PLENUM	270	24		70	12	9
SD-32	Supply diffuser	115	FPB-S1	270	24	8	59	7.5	9
RG-16	Return grille	115	PLENUM	270	24		58	12	9
SD-33	Supply diffuser	116	FPB-S1	270	24	8	50	7.5	9

Tag	Type	Room	Served by	cfm	Face in.	Neck in.	X ft	Y ft	Ceiling ft
RG-17	Return grille	116	PLENUM	270	24		46	12	9
SD-34	Supply diffuser	117	FPB-S2	270	24	8	38	7.5	9
RG-18	Return grille	117	PLENUM	270	24		34	12	9
SD-35	Supply diffuser	118	FPB-S2	270	24	8	26	7.5	9
RG-19	Return grille	118	PLENUM	270	24		22	12	9
SD-36	Supply diffuser	113	FPB-SE	360	24	10	94	14	9
SD-37	Supply diffuser	113	FPB-SE	360	24	10	86	6	9
RG-20	Return grille	113	PLENUM	360	24		86	14	9
RG-21	Return grille	113	PLENUM	360	24		94	6	9
SD-38	Supply diffuser	120	FPB-W1	245	24	8	7.5	26	9
RG-22	Return grille	120	PLENUM	245	24		12	30	9
SD-39	Supply diffuser	121	FPB-W1	245	24	8	7.5	41	9
RG-23	Return grille	121	PLENUM	245	24		12	42	9
SD-40	Supply diffuser	122	FPB-W1	245	24	8	7.5	50	9
RG-24	Return grille	122	PLENUM	245	24		12	54	9
SD-41	Supply diffuser	123	FPB-W2	245	24	8	7.5	62	9
RG-25	Return grille	123	PLENUM	245	24		12	66	9
SD-42	Supply diffuser	124	FPB-W2	245	24	8	7.5	74	9
RG-26	Return grille	124	PLENUM	245	24		12	78	9
SD-43	Supply diffuser	119	FPB-SW	360	24	10	14	6	9
SD-44	Supply diffuser	119	FPB-SW	360	24	10	6	14	9
RG-27	Return grille	119	PLENUM	360	24		14	14	9
RG-28	Return grille	119	PLENUM	360	24		6	6	9
RG-29	Return grille	127	PLENUM	495	24		40	40	9
RG-30	Return grille	128	PLENUM	495	24		60	40	9
RG-31	Return grille	125	PLENUM	495	24		34	74	9
RG-32	Return grille	126	PLENUM	495	24		66	74	9
RG-33	Return grille	127	PLENUM	495	24		26	46	9
RG-34	Return grille	128	PLENUM	495	24		74	46	9

Tag	Type	Room	Served by	cfm	Face in.	Neck in.	X ft	Y ft	Ceiling ft
RG-35	Return grille	128	PLENUM	495	24		50	30	9
RG-36	Return grille	126	PLENUM	495	24		50	78	9

8. Checks performed on the model

The following automated checks ran on the final design and passed. They are geometric and arithmetic checks, not an engineering review.

1. Clash check: every straight duct segment, terminal box, air handler and column is represented by a bounding box; no two overlap except where runs legitimately meet at a node or tap. Interior walls are not included in the check (ducts pass through partitions above the ceiling).
2. Connectivity check: every run that names a piece of equipment starts or ends exactly on that equipment's inlet or outlet connector.
3. Velocity check: no medium-pressure run exceeds 2,000 fpm and no low-pressure run exceeds 1,200 fpm. Actual maxima: 1,472 fpm medium pressure (SA-01 riser), 882 fpm low pressure.
4. Area check: the 29 space polygons sum to 10,000.0 sf with no gaps or overlaps by construction.
5. Airflow accounting: diffuser airflows equal the zone supply airflows within rounding; box primary air equals the sum of the taps on each trunk; trunk sizes are recomputed from the remaining airflow after every tap.
6. IFC validation: the IFC4 file passes the ifcopenshell schema validator with no errors, and every one of the 472 solids rebuilds in the geometry kernel.
7. Terminal spacing: no two air terminals are within 2 ft of each other.

8.1 What was not checked

- Cooling and heating loads, psychrometrics, altitude corrections, coil selections.
- Duct pressure drop, fan static pressure, static regain, acoustics, or the ASHRAE 62.1 multiple-zone outside air calculation.
- Code compliance (IMC, IECC/90.1, 62.1, NFPA 90A), smoke dampers at the mechanical room, fire dampers, plenum-rated materials.
- Structural coordination (duct hangers, roof openings), beam depths (none modeled), and the 1 ft 6 in. clearance assumed between the ceiling and the bottom of the largest duct.
- Revit execution: the Revit API script was written against the 2022 to 2025 API but has not been run in Revit; the IFC was generated and verified in Python.

9. Items for engineering review

These are the points where a professional judgement or calculation should replace the author's assumption.

1. Exposure factors (S 1.50, W 1.35, E 1.25) and the 1.0 cfm/sf north office baseline. A room-by-room load calculation for Denver with the actual glazing will set the real airflows. Conference rooms in particular may need more than 1.2 cfm/sf before the exposure factor, or CO2 demand control.
2. Series versus parallel fan-powered boxes, and the 1.5 fan-to-primary ratio. Parallel boxes with a lower fan airflow may be preferred for energy; series boxes were chosen for constant room airflow and better winter mixing on the perimeter.
3. Electric reheat at 25 Btuh/sf. Confirm the perimeter heating load (Denver 99% design about 0 deg F; 40% glazing) and whether the owner prefers hot water. Check 90.1 reheat and fan-power limits.
4. Outside air: 1,800 cfm was estimated from 62.1 rates and a 0.75 system efficiency. The multiple-zone calculation with the actual critical zone (likely a south conference room at minimum airflow) should be run, and demand-control ventilation considered. Relief and building pressurization are not addressed.

5. AHU rating: 10,000 cfm, 25 tons, 300 MBH, 2.5 in. w.g. ESP are placeholders. The block load, altitude correction (about 0.82 air density at 5,280 ft) and the actual duct pressure drop will set these. The 36x24 riser at about 1,470 fpm and 28x16 mains at about 1,420 fpm are at the high end for a medium-pressure office system; a static-regain sizing and an acoustic check at the first boxes are recommended.
6. Ring main concept: four dead-ended legs fed from the east and west mains rather than a closed loop. A closed loop would give pressure equalization; the dead ends simplify balancing and the Revit system definition.
7. Mechanical room: 24 x 16 ft with a 16 x 6 ft unit and all three risers on top of the unit. A real unit of this size (10,000 cfm) is about 20 ft long with end connections; check the room, door, service access and acoustic separation from the adjacent open office.
8. Return path: every perimeter room has its own ceiling return grille so partitions to the ceiling do not block return. If partitions go to the deck for acoustics, transfer ducts are needed. The plenum must be rated for return air.
9. The middle office of each three-office group has its diffuser 3 ft off the room centre line to clear the header tee. Confirm throw and coverage, or move the box.
10. Outside-air louver L-1 sits above the ceiling in the north wall of North Office 4 (X 58 ft). Check that it clears the window head and is acceptable on that elevation, and that the OA duct can be insulated and sloped.
11. Interior VAV minimums (30%) and whether the interior zones need any heating (none provided).

10. Revision history

Rev	Change
1	Open-plan office, 8 perimeter fan-powered boxes in 20 ft deep perimeter zones (two per facade), 4 interior VAVs, AHU 8,500 cfm, uniform 1.0 to 1.25 cfm/sf, 48 diffusers, 16 return grilles.
2	Perimeter rezoned to 20 private offices and 4 corner conference rooms with a 5 ft corridor; plan north fixed as up; Denver exposure factors applied (S 1.50, W 1.35, E 1.25); 12 perimeter fan-powered boxes (one per conference room, one per 2 to 3 offices); AHU 10,000 cfm with 1,800 cfm OA; ducts resized automatically from airflow along a ring main; OA louver moved from X 47.5 to X 58 to clear the north discharge legs; return duct 48x28; 44 diffusers and 36 return grilles; 44 walls.

11. How the model was built

design.py holds the building, the zones, the equipment, the duct network and the sizing rules in one Python module. From it:

- build_ifc.py writes the IFC4 model with ifcopenshell. Every element is a swept solid placed in feet, typed (duct sizes, box types, wall types) and grouped into Supply Air, Return Air and Outside Air distribution systems, with an HVAC_Design property set on every element (airflow, size, velocity, pressure class, zone).
- CreateOfficeModel.py reads design.json inside Revit and creates native levels, grids, walls, floor, roof, ceilings, spaces with separation lines, ducts with fittings (elbows, transitions, tees, crosses, takeoffs from the duct type's routing preferences), equipment and air terminals, using library families when found and DirectShape placeholders otherwise.
- make_drawings.py runs the clash, connectivity and velocity checks and writes the schedules and the plan.
- render_iso.py draws the axonometric from the IFC geometry, and build_review_doc.cjs writes this document from the same data.

Coordinates: all values in this package are in feet with the origin at the outside south-west corner of the building, X to the east, Y to the north, Z up from Level 1.